

Detection of Sepsis using Machine Learning: A Comprehensive Evaluation of Model Performance

Mallemputi Sai Teja
Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Chennai, India.
mallemputisaiteja9999@gmail.com

Bandaru Rohan Satya Balaji
Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Chennai, India.
rohanbandaru14838@gmail.com

Tunga Vignesh
Computer Science and Engineering
Amrita School of Computing
Amrita Vishwa Vidyapeetham
Chennai, India.
tungavignesh@gmail.com

E Sophiya
Computer Science and Engineering
Vel Tech Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
Chennai, India.
venus.sophiya@gmail.com

Abstract—Sepsis is a life-threatening condition arising from the body's response to infection with high mortality and escalating race, representing one of healthcare's remaining challenges. Prediction of sepsis at an early stage leads to patient safety and also helps reduce the cost of healthcare. This paper is focused on investigating the performance of different machine learning (ML) based models for early sepsis detection, given a dataset from Computing in Cardiology Challenge. The data was preprocessed, normalized, and features were selected for model training. Models including Decision Tree, Random Forest, Extra Trees, XGBoost. Results indicate that the Extra Trees classifier provided the highest accuracy, precision and recall (99.7%, 94.0% and 90.0% respectively). The results of this study highlight the potential of ML techniques to improve early sepsis diagnosis, which can result in prompt medical attention and improved patient outcomes.

Index Terms—Sepsis, Early Detection, Machine Learning, Random Forest, Extra Trees Classifier, Healthcare Analytics, Predictive Modeling, Class Imbalance, Data Preprocessing, Feature Selection

I. INTRODUCTION

Sepsis is a condition that occurs when the body overreacts to an infection leading to a potentially life threatening situation. Due, to its high mortality rate and rapid progression sepsis poses challenges, for hospitals. Detecting sepsis on can not reduce healthcare expenses but also enhance patient well being and safety.

While sepsis can affect anyone some individuals are, at risk such as the young children, expectant mothers and those, with preexisting health conditions. Common signs of sepsis consist of fever, fast breathing, elevated heart rate, confusion and general unease. Neglecting sepsis can result in fatal outcomes such as multiple organ failure, septic shock, and even death. Most often, bacterial infections are the cause of sepsis. Still, fungal, viral, or parasite infections can also cause it. Treatment usually involves intravenous fluids, antibiotics, and other supportive measures urgent medical intervention is required. [1].

In India, there were around 3 million deaths and an estimated 11 million cases of sepsis in 2017. As per a 2018 study published in the Lancet Journal of Respiratory Medicine sepsis was a factor in 20.3% of deaths in England. Based on the estimated 245,000 instances of sepsis that occur in the UK each year, extrapolating this data indicates that 49,735 lives are lost annually.

CDC says that, sepsis is an immune response illness that affects over 1.7 million people in the US each year. Surprisingly, in American hospitals, it ranks as the third most common cause of mortality. Furthermore, sepsis is the most expensive medical condition; an estimated \$62 billion is spent annually on hospital stays and skilled nursing care.

One of the leading causes of death globally is sepsis. According to data published in 2020 [2], there were 48.9 million cases of sepsis worldwide, leading to 11 million deaths related to sepsis, approximately 20% of all deaths worldwide. Additionally, there are 24 million pediatric cases of sepsis annually, which lead to 2.9 million deaths. With early identification and appropriate clinical care, the majority of these deaths might be prevented.

About 15 patients out of every thousand hospitalized patients might get sepsis as a side effect of their treatment. Additionally, sepsis is extremely pricey. In high-income nations, the average hospital treatment cost for sepsis is estimated to be more than \$32,000 per patient.

Given the seriousness and prevalence of sepsis, developing a strong early detection system could greatly help medical professionals intervene in a timely manner, possibly saving lives. This study emphasizes the need for prediction algorithms that make use of clinical data to enhance patient prognosis and sepsis detection. Healthcare systems can proactively identify persons at risk of this dangerous illness and minimize mortality rates and associated healthcare costs by utilizing advanced analytics and machine learning approaches.

This study utilizes a dataset from the Computing in Cardiology Challenge, focusing on early sepsis prediction from clinical data, which includes demographics, vital signs, and laboratory values. Effective data preprocessing, including handling missing values, normalization, and class balancing with techniques like SMOTE, is crucial for enhancing predictive modeling. Various machine learning were evaluated to determine the most effective method for early sepsis detection.

II. LITERATURE SURVEY

V. R. Burugadda et al. [3] proposed an ML system with gradient boosting to detect sepsis early and achieved an accuracy of 92%. This system makes it easier to render quick clinical decisions but needs to be tested in many different environments of healthcare. S. Lyra et al. [4] proposed an ML models for early neonatal sepsis detection, addressing its rapid progression and non-specific symptoms. Four models are optimized and evaluated using both adult and neonatal clinical data, with synthetic data augmentation to tackle the limited neonatal sepsis data. The results are promising, with an AUROC of 0.91 and AUPRC of 0.38, suggesting ML's potential for early sepsis prediction.

S. K. Rout et.al [5] discusses ICU early sepsis detection by the use of ML, which attained a mean AUROC of 0.696. SVM is accurate at predicting the outcome of severe sepsis, and LSTM shows 96% training and 95% validation accuracy using PhysioNet Challenge 2019 data.

D. Bhaskaracharya and D. Mehta [6] conducted another study in which they used machine learning models to diagnose early sepsis using data from the PhysioNet 2019 sepsis challenge. In this diagnostic, the Random Forest classifier achieves 96% accuracy.

M. R. Kumar,et al. in their study [7] shows how critical it is to detect sepsis at an early stage, focusing on how effective machine learning algorithms can be. It presents the ensemble learning of Random Forest and LSTM models on a dataset of 10,000 patients, with an accuracy of 95.5% and an AUC of 0.95 for Random Forest, and an accuracy of 88.6% and an AUC of 0.94 for LSTM. Combining those models enhances accuracy to 95.5% and AUC to 0.98. Validation on the PhysioNet Challenge 2019 dataset proves that the ensemble approach is really efficient; it underlines the fact that the model might enhance early sepsis detection and patient outcomes. S. N et al., in their study [8], identified the critical need for the early detection of sepsis and proposed the development of a machine learning-based model by using vital signs. It has achieved an accuracy of 98% and hence indicates the potential for timely intervention in low- and middle-income countries.

S. Aref et al. [9] presented an early sepsis detection segmentation methodology using a large clinical dataset. XGBoost yielded a recall of 0.98 and AUROC of 0.98 six hours prior to onset, demonstrating time series data segmentation and feature engineering.

M. Ashikur Rahman, A. AbuBakar Ibrahim and A. Tumian, in their study [10] emphasizes the use of EHR data for early detection of sepsis in ICUs using machine learning models.

Feature selection based on GLM enhances model accuracy. Compared to the clinical scores, the accuracy of the RF model reached 90.1%, while the sensitivity was as high as 99.49% in another paper. The authors [11] discuss the potential of AI in clinical decision support for sepsis, referencing work done in this area that has focused on predicting, identifying, sub-phenotyping, prognosticating, and treating sepsis with AI. They also address some of the challenges facing the development of such tools, such as the need for further testing, privacy, and integration with healthcare IT. The authors conclude that, while promising, there are significant obstacles to the widespread adoption of AI-based modeling tools in clinical care.

The authors A. Honore' et al. in their study [12] stresses the importance of the timely identification of neonatal sepsis for the reduction of mortality and morbidity in infants born prematurely. It examines the comparison, between predictors and recurrent neural networks particularly focusing on long short term memory structures using real time data, from the Philips IntelliVue MX800 Patient Monitor. In another The authors in the study [13] underlines the importance of early detection of sepsis because of its high mortality rate and discusses machine learning algorithms such as Random Forest and Logistic Regression. Techniques include Principal Component Analysis and SMOTE. The model using Logistic Regression and SVM resulted in 98% accuracy using PhysioNet Challenge 2019 data.

J. Xun et al. in their study [14] emphasizes the importance of early sepsis detection and how using machine learning algorithms can improve patient outcomes. Approaches such as the STraTS algorithm for time-series forecasting showed high accuracy in sepsis prediction, utilizing 24-hour patient data in a data-driven setup.

K. M and V. S. Akshaya in their study [15] underscores the critical importance of early detection of sepsis and uses an XG-Boost algorithm for its prediction, achieving 95.01% accuracy. It outperforms models such as Decision Tree, Gradient Boosting Tree and Random Forest and suggests further refinement with domain experts and possibly the use of deep learning.

The literature review highlights the encouraging performance of ML and DL models in sepsis detection. Nonetheless, significant opportunities remain to further enhance these models' effectiveness, particularly in intricate clinical scenarios. Additionally, integrating diverse data sources and refining real-time analysis capabilities could further improve sepsis detection accuracy.

III. METHODOLOGY

The Proposed methodology for sepsis detection is involved in various preprocessing steps which is shown in Figure 1.

A. Dataset

The data used in this research comes from the computing, in cardiology challenge known as prediction of sepsis from data [16] [17]. The dataset is divided into three categories; Demographics, Vital Signs and Laboratory values. Out of the

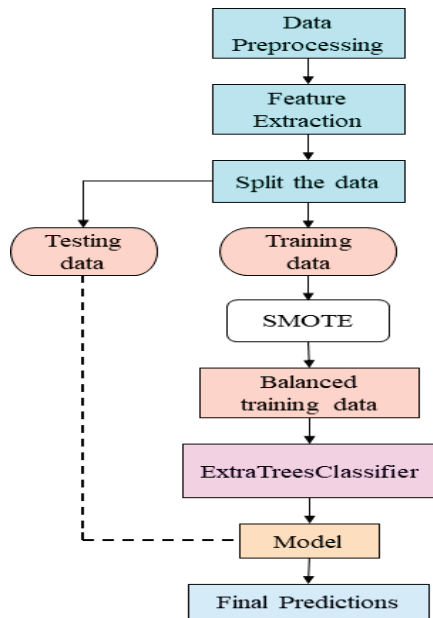


Fig. 1. Proposed Methodology

40 attributes 8 are related to signs, 6 pertain to Demographics and the rest fall under Laboratory values. Demographics cover factors like Age, Gender and the time between hospital admission and ICU admission. Laboratory values include parameters such as Platelets, Fibrinogen, White Blood Cell count (WBC) and Hemoglobin (Hgb). Vital signs encompass metrics like Heart rate Pulse oximetry levels, Temperature readings and Respiration rate. The target feature SepsisLabel indicates whether a patient has sepsis with a value of 1, for patients and 0 for septic patients.

B. Data Preprocessing

1) **Data Cleaning:** The dataset contains missing values. Attributes with significant absence of data were identified. Some attributes like Bilirubin_direct, Fibrinogen, and TroponinI have more than 99% of null values.

For enhancing the dataset's relevance and to get rid of redundancy, certain features were removed. The columns Unnamed: 0, SBP, DBP, EtCO2, BaseExcess, HCO3, pH, PaCO2, Alkalinephos, Calcium, Magnesium, Phosphate, Potassium, PTT, and Fibrinogen were dropped. The availability of data, redundancy, focus on more insightful health indicators and the readily accessible nature of data established the foundation for these decisions.

After the initial cleaning steps to address remaining missing values forward and backward filling were used within patient groups to efficiently propagate values. This imputation technique has helped maintain the consistency of the time series data. The features with more than 35% null values along with

patient id were removed after applying the imputation strategy. Finally, all the instances having null values were excluded.

2) **Data Transformation:** Normalization and Log Transformation were selectively applied to the features. Log transformations were used for features like MAP, BUN, Creatinine, Glucose, WBC, and Platelets. Because these values can vary significantly and applying log transformation will help us to stabilize variance and handle skewed distributions.

Standard normalization was performed on a subset of features, including HR, O2Sat, Temp, MAP, Resp, BUN, Chloride, Creatinine, Glucose, Hct, Hgb, WBC, and Platelets.

The Gender attribute undergone one-hot encoding as there is no inherent ordinal relationship between the categories.

C. Feature Selection

After deleting the attributes which were non relevant and having more null values the total number of features became 19 including the target variable. The information about the features in the final data is given the illustrated in Table 1.

TABLE I
DESCRIPTION OF FEATURES

Category	Feature	Description
Vital signs	HR	Heart rate (beats per minute)
	O2Sat	Pulse oximetry (%)
	Temp	Temperature (Deg C)
	MAP	Mean arterial pressure (mm Hg)
	Resp	Respiration rate (breaths per minute)
Laboratory value	BUN	Blood urea nitrogen (mg/dL)
	Chloride	(mmol/L)
	Creatine	(mg/dL)
	Glucose	Serum glucose (mg/dL)
	Hct	Hematocrit (%)
	Hgb	Hemoglobin (g/dL)
	WBC	(count*10 ³ /μL)
Demographics	Platelets	(count*10 ³ /μL)
	Age	Years (100 for patients 90 or above)
	HospAdmTime	Hours between hospital admit and ICU admit
	ICULOS	ICU length-of-stay (hours since ICU admit)
Outcome	Gender	Female (0) or Male (1)
	SepsisLabel	For sepsis patients, SepsisLabel is 1

D. split the data

In the sci kit learn library, the train_test_split() function, in the model selection module is commonly employed to divide data into training and testing subsets. Typically 80% of the data is assigned for training with the remaining 20% set aside for testing purposes.

E. Class Balancing

The dataset contains 98% instances without sepsis and about 2% of instances with sepsis which is represented in Figure 2. In order to balance the samples between two classes, oversampling technique was applied. Class imbalances occur when the

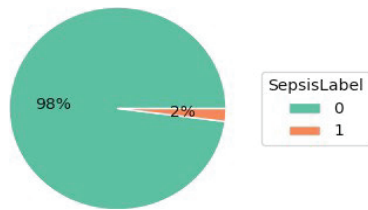


Fig. 2. Distribution of the dataset

minority class is significantly outnumbered by the majority class leading to models that struggle to predict the minority class. To address this issue oversampling entails duplicating or generating instances of the minority class randomly thus achieving a balanced distribution, within the dataset.

SMOTE (Synthetic Minority Over-sampling approach) is a common oversampling approach used in machine learning datasets to address class imbalance by artificially producing instances of the minority class. Finding examples in the dataset that fall into the minority class, which is underrepresented in comparison to the dominant class is the first step in the process. To balance the distribution of classes, SMOTE creates synthetic instances along the line segments that connect each minority class instance to one or more of its closest neighbors that are also members of that class. Typically the process involves picking a spot, along the line connecting a minority class data point and one of its neighboring points. SMOTE works by generating samples to boost the number of minority class instances in the dataset. This helps balance out the distribution of classes making the dataset more effective, for training machine learning models. Figure 3 illustrates how SMOTE approach has efficiently balanced the samples between sepsis and non sepsis.

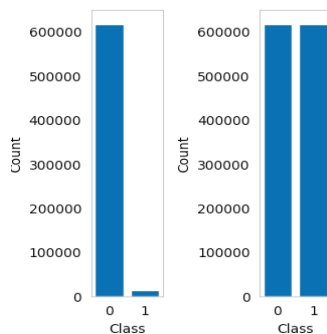


Fig. 3. a) training data before balancing. b) training data after balancing.

F. Model Training & Evaluation

1) **Decision Tree:** Decision Trees serve in handling both classification and regression problems. Decision Tree is a

supervised learning algorithm. They construct a tree-shaped framework, enabling decision-making through iterative division of data, predicated on distinctive features. The model was trained with the gini index criterion to identify the best features from the dataset and a max_depth of 3 which is shown in Figure 4. By utilizing Decision Tree, the accuracy achieved was 97.8%, the precision was 46.4%, and the recall was 60.6%.

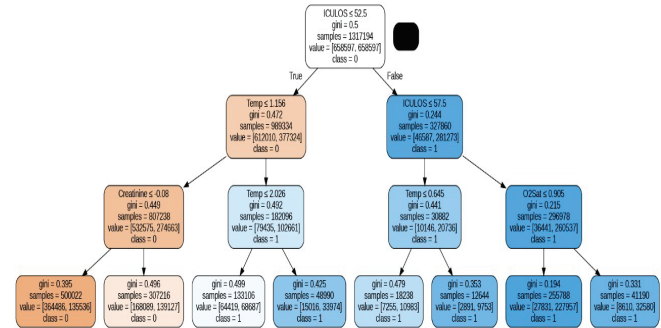


Fig. 4. Decision Tree for the Trained Model

2) **Random Forest:** By utilizing an ensemble of decision trees that incorporate a variety of clinical features, this method is able to generate accurate predictions. Through the process of voting, each decision tree contributes to the overall prediction, leading to a stable and reliable detection system that is capable of capturing intricate patterns associated with early sepsis. By combining the results of numerous trees, Random Forest is able to provide a well-balanced and precise approach to early sepsis identification. Its ability to adapt and effectively analyze a diverse range of patient data truly sets Random Forest apart as a top choice for sepsis detection. With the Random Forest machine learning model, 99.5% accuracy, 92.1% precision, and 83.2% recall were achieved.

3) **XGBoost:** XGBoost, a powerful ensemble learning algorithm based on the boosting concept, effectively detects sepsis in patient data. By using decision trees as the building blocks of its model, XGBoost continuously improves its performance by optimizing an objective function through the use of gradient descent. Additionally, XGBoost employs the technique of early stopping to protect against overfitting and utilizes parallel and distributed computing to improve its efficiency. This robust approach makes XGBoost an invaluable tool in accurately identifying sepsis in patient data. The XGBoost machine learning model achieved an accuracy of 96.6%, precision of 27.7%, and recall of 40.3%.

4) **Extra Trees Classifier:** Extra Trees utilizes additional randomness in building decision trees to extract a wide range of clinical features that are crucial in early detection of sepsis. This method prevents overfitting and promotes generalization by constructing multiple trees with increased variability. By examining random sections of patient data, Extra Trees effectively captures subtle patterns that can signal the onset of sepsis. Therefore, it effectively leverages enhanced

TABLE III
COMPARITIVE ANALYSIS OF EXISTING MODELS

Existing Models	Accuracy	Existing Model Accuracy
Decision Tree [15]	97.8%	91.6%
Random Forest [6]	99.5%	96.0%
XGBoost [15]	96.6%	95.0%

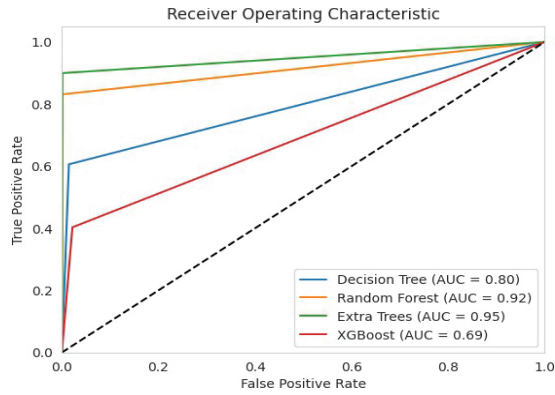


Fig. 7. Evaluation using ROC

V. CONCLUSION

This study highlights the critical need for early sepsis detection and demonstrates the efficacy of various machine learning models in achieving this goal. With an f1 score of 92%, the Extra Trees classifier proved to be the most successful of the models tested. This indicates that the model can handle unbalanced data and identify complex patterns linked to the start of sepsis. The study also shown how crucial careful feature selection and data preprocessing are to enhancing model performance.

The use of advanced analytics and machine learning techniques holds significant promise for reducing sepsis-related mortality and healthcare costs; however, future work should concentrate on real-world validation, integration with clinical workflows, and addressing challenges related to data privacy and system interoperability.

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